A Single Snapshot DOA Estimation Method Based on ADMM-Net

Jiachen Wang, Yifeng Wu, *Member, IEEE*, Xiaobo Deng, Lei Zhang, *Member, IEEE*, Dong Xia, and Qinquan Zhou

***Abstract*— Direction of arrival (DOA) can be estimated through sparse recovery (SR) methods based on the sparsity of signals. However, conventional SR-DOA methods, such as the alternating direction method of multipliers (ADMM), encounter issues such as difficulty in parameter setting and insufficient estimation accuracy. ADMM-Net is a neural network that combines ADMM with deep unfolding (DU). In this letter, we propose a method for estimating DOA using ADMM-Net. The simulation results demonstrate that the proposed method achieves a better balance between DOA estimation performance and real-time constraints.**

***Index Terms*— Alternating direction method of multipliers (ADMM), deep learning (DL), direction-of-arrival (DOA) estimation, sparse recovery (SR).**

I. Introduction

DIRECTION-OF-ARRIVAL (DOA) estimation is widely applied in various fields such as radar, sonar, electronic surveillance, and wireless communications. Typical super-resolution DOA estimation methods include the multiple signal classification (MUSIC) [1] and the estimation of signal parameters via the rotational invariance technique (ESPRIT) [2]. These methods can achieve high angular resolution, but they require the estimation of the signal covariance matrix using multiple snapshots of received data. However, in practical applications, it is difficult for the received data to meet the requirement. To address these issues, researchers have proposed improved methods such as spatial smoothing [3], using even a single snapshot of received data to estimate the signal covariance matrix for DOA estimation. However, these methods often suffer from problems such as reduced angular resolution and increased computational complexity.

In recent years, there have been many new DOA estimation methods based on the sparse recovery (SR) theory. By dividing the angular space into grids and constructing an overcomplete dictionary of steering vectors, the received data exhibit sparsity. This allows the DOA estimation problem transform into an SR problem. The algorithms used to solve the SR-DOA problems include orthogonal matching pursuit (OMP) algorithm [4], smoothed $L0$ (SL0) algorithm [5], and $L1$-norm minimization algorithm [6]. Among them, the OMP algorithm has a low computational cost and estimation error but typically relies on prior information such as signal sparsity. The SL0 algorithm has low computational complexity, but it requires parameter tuning. The $L1$-norm minimization algorithm transforms the SR problem into a second-order cone programming problem. An interior-point method is traditionally used to solve this problem, providing high estimation accuracy but with huge computational cost. To overcome the issue of computational cost, Wang et al. [7] introduce an alternating direction method of multipliers (ADMM) to solve this problem. ADMM guarantees convergence while decreasing computational complexity, but it commonly takes hundreds of iterations to converge and requires parameter tuning.

With the development of deep learning (DL), some researchers apply deep neural networks (DNNs) to DOA estimation. Zhao et al. [8] transform the DOA estimation problem into a multiclassification problem and implement the DOA estimation using convolutional neural networks (CNNs). Elbir [9] proposes the Deep-MUSIC method, which employs CNN to learn the relationship between the covariance matrix and the MUSIC spatial spectrum. Ma et al. [10] introduce spatial filtering with pre-rotation before using CNN for DOA estimation, resulting in improved DOA estimation performance.

Neural network-based methods are often challenged in interpretability. To leverage the advantages of DNN while maintaining the interpretability of algorithms, Gregor and LeCun [11] propose the deep unfolding (DU) method. The DU method unfolds a specific iterative algorithm into a DNN. The parameters of DNN represent the iterative parameters of the algorithm. Zhu et al. [12] propose learned modified SL0 (LMSL0) by unfolding modified SL0. To address the parameter tuning problem of ADMM and achieve a better balance between estimation performance and real-time constraints, in this letter, we propose ADMM-Net for SR-DOA. The ADMM-Net can acquire a set of suitable parameters through training, which need to be manually adjusted in the conventional ADMM algorithm. The simulation results demonstrate that the proposed method has achieved a better balance between DOA estimation performance and real-time constraints compared to the iterative adaptive approach for amplitude and phase estimation (IAA-APES) [13], CVX [14], ADMM, and LMSL0.

Received 15 January 2024; revised 24 April 2024; accepted 15 September 2024. Date of publication 3 October 2024; date of current version 10 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62201623, Grant 62101603, and Grant 62371446; in part by Shenzhen Science and Technology Program under Grant KQTD20190929172704911; and in part by the Insight Action under Grant 62502010220. *(Corresponding author: Yifeng Wu.)*

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Digital Object Identifier 10.1109/LGRS.2024.3469971

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II. Signal Model

A. SR-DOA Model

Consider K narrowband far-field sources and their signals impinging on a uniform linear array (ULA) composed of M elements. Since the time delay of the signal between different elements can be expressed as phase shifts, the received data can be represented as

$$\boldsymbol{y} = \sum_{k=1}^{K} \boldsymbol{a}(\theta_k) s_k + \boldsymbol{n} \tag{1}$$

where s_k is the amplitude of the kth signal and $\boldsymbol{n} \in \mathbf{C}^{M\times 1}$ denotes the Gaussian white noise. $\boldsymbol{a}(\theta_k)$ is the steering vector of the kth signal, represented as

$$\boldsymbol{a}(\theta_k) = \left[1, e^{j2\pi d \sin\theta_k/\lambda_l}, \ldots, e^{j2\pi(M-1)d\sin\theta_k/\lambda_l}\right]^{\mathrm{T}} \tag{2}$$

where d, λ_l, and θ_k denote the interelement distance, the radar operating wavelength, and the DOA of the kth signal, respectively; $d\sin\theta_k/\lambda_l \in [-1/2, 1/2]$ is the spatial frequency; and $[\cdot]^{\mathrm{T}}$ represents the matrix transposition.

More compactly, (1) can be written as

$$\boldsymbol{y} = \boldsymbol{A}_K(\theta)\boldsymbol{S} + \boldsymbol{n} \tag{3}$$

where $\boldsymbol{A}_K(\theta) = [\boldsymbol{a}(\theta_1), \boldsymbol{a}(\theta_2), \ldots, \boldsymbol{a}(\theta_K)] \in \mathbf{C}^{M\times K}$ is the array manifold matrix and $\boldsymbol{S} = [s_1, s_2, \ldots, s_K] \in \mathbf{C}^{K\times 1}$ is the amplitude vector of signals. The DOA estimation is to estimate the DOAs $\theta_1, \theta_2, \ldots, \theta_K$ from $\boldsymbol{y}$.

To estimate DOAs based on the SR-DOA model, we make use of the overcomplete dictionary to represent $\boldsymbol{y}(t)$. A dictionary $\boldsymbol{A} \in \mathbf{C}^{M\times N}$ can be constructed based on potential DOAs $\theta_1, \theta_2, \ldots, \theta_N$ in the region of interest, where N is the number of potential DOAs and $N \gg K$. Then, the received data can be represented as

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{s} + \boldsymbol{n} \tag{4}$$

where $\boldsymbol{s} \in \boldsymbol{C}^{N\times 1}$ denotes the amplitude vector of N DOAs. Since the number of signals K is typically smaller than the grid size N, $\boldsymbol{s}$ only contains a small number of nonzero elements, indicating that $\boldsymbol{s}$ is sparse. Then, the DOA estimation problem can be transformed into the following optimization problem:

$$\arg\min_{\boldsymbol{s}} \|\boldsymbol{s}\|_0, \quad \text{s.t. } \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{s}\|_2 \le \varepsilon \tag{5}$$

where $\varepsilon \ge \|\boldsymbol{n}\|_2$ is an upper bound on the noise energy and $\|\cdot\|_0$ and $\|\cdot\|_2$ denote the ℓ_0 norm and ℓ_2 norm of a vector, respectively. The ℓ_0 optimization problem in (5) is NP-hard to solve [15], [16]. Therefore, in this letter, we replace the ℓ_0 norm by its tightest convex relaxation—the ℓ_1 norm. Then, (5) is transformed into the following equation:

$$\arg\min_{\boldsymbol{s}} \lambda\|\boldsymbol{s}\|_1 + \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{s}\|_2^2 \tag{6}$$

where λ is the regularization parameter.

B. ADMM Algorithm

The ℓ_1 optimization problem, as shown in (6), is a convex optimization problem that can be solved in polynomial time. This problem can be directly solved using CVX [14]. However, the computational cost of CVX is too heavy, which seriously restricts the application of such DOA estimation methods in engineering. ADMM is a classical convex optimization method that can significantly improve the efficiency of solving convex optimization problems. We next present the solution process of ADMM for the optimization problem shown in (6).

We first introduce an intermediate variable $\boldsymbol{z}$ and construct the following optimization model:

$$\begin{cases} \arg\min\limits_{\boldsymbol{s},\boldsymbol{z}} \lambda\|\boldsymbol{z}\|_1 + \dfrac{1}{2}\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{s}\|_2^2 \\ \text{s.t. } \boldsymbol{s} = \boldsymbol{z}. \end{cases} \tag{7}$$

For this optimization problem, the augmented Lagrange multiplier method can be used for solving. First, construct an augmented Lagrange function as follows:

$$L_\rho(\boldsymbol{s}, \boldsymbol{z}, \boldsymbol{\alpha}) = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{s}\|_2^2 + \lambda\|\boldsymbol{z}\|_1 + \langle \boldsymbol{\alpha}, \boldsymbol{s} - \boldsymbol{z}\rangle + \frac{\rho}{2}\|\boldsymbol{s} - \boldsymbol{z}\|_2^2 \tag{8}$$

where $\boldsymbol{\alpha}$ and ρ are the Lagrange multiplier and penalty parameter, respectively, and $\langle\cdot\rangle$ denotes the inner product. For simplicity, we use a scaled definition $\boldsymbol{u} = (\boldsymbol{\alpha}/\rho)$. Then, the augmented Lagrange function becomes

$$L_\rho(\boldsymbol{s}, \boldsymbol{z}, \boldsymbol{u}) = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{s}\|_2^2 + \lambda\|\boldsymbol{z}\|_1 + \frac{\rho}{2}\|\boldsymbol{s} - \boldsymbol{z} + \boldsymbol{u}\|_2^2 - \frac{\rho}{2}\|\boldsymbol{u}\|_2^2. \tag{9}$$

The ADMM algorithm solves the optimization problem by iteratively solving the following subproblems alternately:

$$\begin{cases} \boldsymbol{s}^{(i)} = \arg\min\limits_{\boldsymbol{s}} L_\rho\left(\boldsymbol{s}, \boldsymbol{z}^{(i-1)}, \boldsymbol{u}^{(i-1)}\right) \\ \boldsymbol{z}^{(i)} = \arg\min\limits_{\boldsymbol{z}} L_\rho\left(\boldsymbol{s}^{(i)}, \boldsymbol{z}, \boldsymbol{u}^{(i-1)}\right) \\ \boldsymbol{u}^{(i)} = \boldsymbol{u}^{(i-1)} + \mu\left(\boldsymbol{s}^{(i)} - \boldsymbol{z}^{(i)}\right) \end{cases} \tag{10}$$

where $i \in [1, 2, \ldots, I_s]$ denotes the ith iteration and μ is an update rate. After substituting (9) into (10), the solutions to the three subproblems are obtained as

$$\begin{cases} \boldsymbol{s}^{(i)} = \left(\boldsymbol{A}^H\boldsymbol{A} + \rho\boldsymbol{I}\right)^{-1}\left[\boldsymbol{A}^H\boldsymbol{y} + \rho\left(\boldsymbol{z}^{(i-1)} - \boldsymbol{u}^{(i-1)}\right)\right] \\ \boldsymbol{z}^{(i)} = S_{\frac{\lambda}{\rho}}\left(\boldsymbol{s}^{(i)} + \boldsymbol{u}^{(i-1)}\right) \\ \boldsymbol{u}^{(i)} = \boldsymbol{u}^{(i-1)} + \mu\left(\boldsymbol{s}^{(i)} - \boldsymbol{z}^{(i)}\right) \end{cases} \tag{11}$$

where $\boldsymbol{I}$ represents the identity matrix and $S_{(\cdot)}(\cdot)$ denotes the soft thresholding operator. For complex-valued data x and real-valued scalar threshold ξ, $S_\xi(x) = (x/|x|)\max(|x| - \xi, 0)$

III. ADMM-Net for DOA Estimation

The proposed ADMM-Net is constructed by DU the ADMM solver. ADMM-Net considers the parameters that require manual setting in ADMM as learnable parameters, enabling it to achieve better performance.

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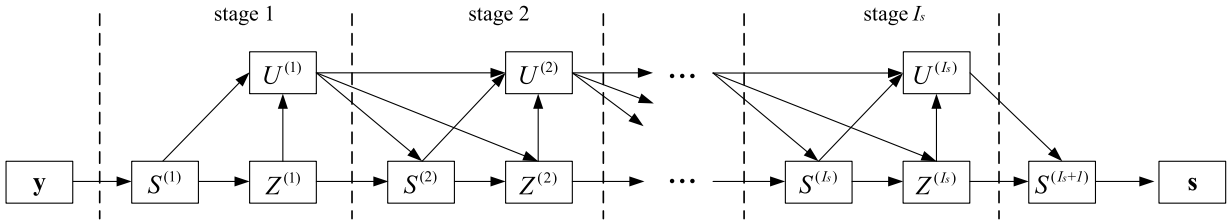

Fig. 1. Data flow graph of ADMM-Net.

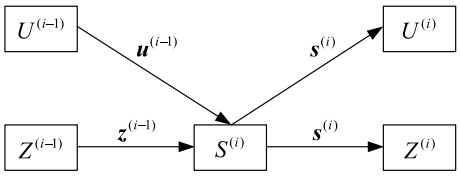

Fig. 2. Data flow graph of reconstruction layer in stage i.

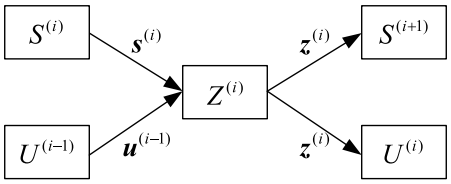

Fig. 3. Data flow graph of nonlinear transform layer in stage i.

A. ADMM-Net

The data flow graph of ADMM-Net is shown in Fig. 1. $S^{(i)}$, $Z^{(i)}$, and $U^{(i)}$ represent the reconstruction layer, nonlinear transform layer, and multiplier update layer of the ith stage, respectively. We next discuss ADMM-Net in detail.

1) Reconstruction Layer: $S^{(i)}$ reconstructs the amplitude vector $\boldsymbol{s}^{(i)}$ according to (11). For $i = 1, 2, \ldots, I_s$, given $\boldsymbol{z}^{(i-1)}$ and $\boldsymbol{u}^{(i-1)}$, the reconstruction layer of ADMM-Net is written as follows:

$$\boldsymbol{s}^{(i)} = \left(\boldsymbol{A}^H\boldsymbol{A} + \rho^{(i)}\boldsymbol{I}\right)^{-1}\left[\boldsymbol{A}^H\boldsymbol{y} + \rho^{(i)}\left(\boldsymbol{z}^{(i-1)} - \boldsymbol{u}^{(i-1)}\right)\right] \quad (12)$$

where $\rho^{(i)}$ is the learnable parameter of the ith stage. The data flow graph of reconstruction layer is shown in Fig. 2. Note that, in the first stage, $\boldsymbol{z}^{(0)}$ and $\boldsymbol{u}^{(0)}$ are initialized to zeros; therefore, $\boldsymbol{s}^{(i)} = (\boldsymbol{A}^H\boldsymbol{A} + \rho^{(i)}\boldsymbol{I})^{-1}\boldsymbol{A}^H\boldsymbol{y}$.

2) Nonlinear Transform Layer: The nonlinear transform layer performs the soft thresholding function in (11). Given $\boldsymbol{s}^{(i)}$ and $\boldsymbol{u}^{(i-1)}$, the output of this layer in stage i is defined as

$$\boldsymbol{z}^{(i)} = S_{\xi^{(i)}}\left(\boldsymbol{s}^{(i)} + \boldsymbol{u}^{(i-1)}\right) \quad (13)$$

where $\xi^{(i)}$ are the learnable parameters of the ith stage. The data flow graph of nonlinear transform layer is shown in Fig. 3. In the first stage, $\boldsymbol{u}^{(0)}$ is initialized to zero, and therefore, $\boldsymbol{z}^{(1)} = S_{\xi^{(1)}}(\boldsymbol{s}^{(1)})$.

3) Multiplier Update Layer: This layer is defined by the Lagrange multiplier updating procedure in (11). Given $\boldsymbol{s}^{(i)}$, $\boldsymbol{z}^{(i)}$, and $\boldsymbol{u}^{(i-1)}$, the output of this layer is defined as

$$\boldsymbol{u}^{(i)} = \boldsymbol{u}^{(i-1)} + \mu^{(i)}\left(\boldsymbol{s}^{(i)} - \boldsymbol{z}^{(i)}\right) \quad (14)$$

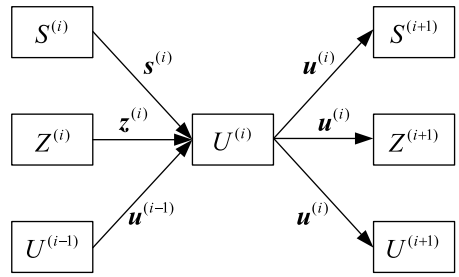

Fig. 4. Data flow graph of multiplier update layer in stage i.

where $\mu^{(i)}$ is the learnable parameter. The data flow graph of multiplier update layer is shown in Fig. 4. In the first stage, $\boldsymbol{u}^{(0)}$ is initialized to zero, and therefore, $\boldsymbol{u}^{(1)} = \mu^{(1)}(\boldsymbol{s}^{(1)} - \boldsymbol{z}^{(1)})$.

In the deep architecture of ADMM-Net, we aim to learn the following parameters:

$$\boldsymbol{\Theta} = \left\{\rho^{(i)}, \xi^{(i)}, \mu^{(i)}\right\}_{i=1}^{I_s}. \quad (15)$$

Compared to the manually set parameters $\{\rho, \lambda, \mu\}$ in the ADMM algorithm, ADMM-Net has more learnable parameters. This gives ADMM-Net greater flexibility and representational capacity compared to the ADMM algorithm and it reduces iterations.

B. Network Training and Backpropagation

We use the received data $\boldsymbol{y}$ as the input of ADMM-Net, and the angle-domain complex-valued vector $\boldsymbol{s}^{\mathrm{gt}}$, which contains the amplitude information of the true signals, as the ground truth. These form the training set $\Gamma = \{\boldsymbol{y}, \boldsymbol{s}^{\mathrm{gt}}\}$. $\hat{\boldsymbol{s}}(\boldsymbol{y}, \boldsymbol{\Theta})$ represents the network output corresponding to the network input $\boldsymbol{y}$ under the network parameter $\boldsymbol{\Theta}$. We choose mean square error (mse) as the loss function in network training, which can be represented as

$$E(\boldsymbol{\Theta}) = \frac{1}{N_c}\sum_{(\boldsymbol{y},\boldsymbol{s}^{\mathrm{gt}})\in\Gamma}\frac{\|\hat{\boldsymbol{s}}(\boldsymbol{y}, \boldsymbol{\Theta}) - \boldsymbol{s}^{\mathrm{gt}}\|_2^2}{N} \quad (16)$$

where N_c represents the total number of samples in the training set and N is the number of potential DOAs. We optimize the parameters of ADMM-Net using gradient-based algorithm Adam [17]. Thus, we need to compute the gradients using backpropagation over the architectures. In this letter, the gradients can be calculated by the following formulations:

$$\begin{aligned}\frac{\partial E}{\partial \boldsymbol{\Theta}_i} &= \Re\left(\frac{\partial E}{\partial O_i}\right)\Re\left(\frac{\partial O_i}{\partial \boldsymbol{\Theta}_i}\right) + \Im\left(\frac{\partial E}{\partial O_i}\right)\Im\left(\frac{\partial O_i}{\partial \boldsymbol{\Theta}_i}\right)\\ \frac{\partial E}{\partial I_i} &= \Re\left(\frac{\partial E}{\partial O_i}\right)\Re\left(\frac{\partial O_i}{\partial I_i}\right) + j\Im\left(\frac{\partial E}{\partial O_i}\right)\Im\left(\frac{\partial O_i}{\partial I_i}\right)\end{aligned} \quad (17)$$

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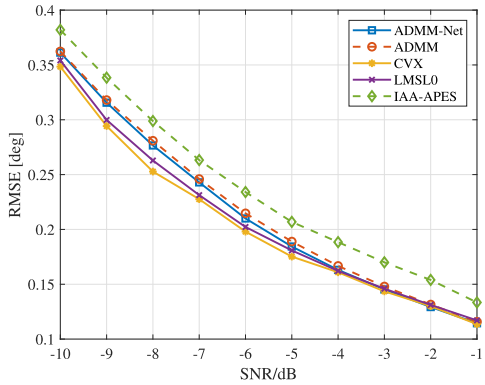

Fig. 5. RMSE performance for the one-source case.

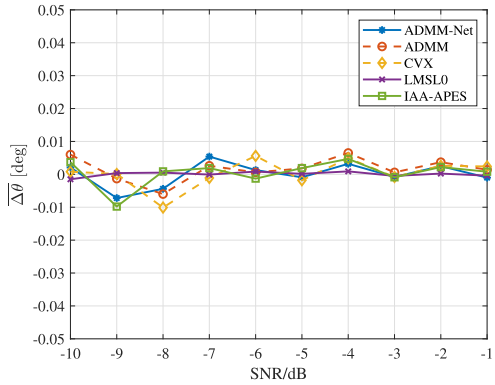

Fig. 6. $\overline{\Delta\theta}$ performance for the one-source case.

where $\mathbf{\Theta}_i$ is the parameters of the ith layer; $\Re$ and $\Im$ are the real and imaginary parts of a formula, respectively; and I_i and O_i represent the input and output of this layer, respectively.

IV. Simulation Results

Consider a ULA with 64 elements with the interelement distance $d = \lambda_l/2$. The number of potential DOAs is $N = 720$. The stage number of ADMM-Net is $I_s = 60$. The signal-to-noise ratio (SNR) of samples in the training set is 20 dB. The number of samples in the training set $N_c = 5 \times 10^3$. We set the learning rate to 0.0005 and use cosine annealing to adjust the learning rate. To verify the superiority of the proposed method in DOA estimation, we compare the proposed ADMM-Net with LMSL0, IAA-APES, CVX, and ADMM algorithm. A root-mean-squared error (RMSE) is used to measure the DOA estimation performance. We also perform a run-time analysis at the end of this section.

A. One-Source DOA Estimation Performance Comparison

In this section, we examine the estimation performance versus SNR for one source. The SNR of the test set in this part ranges from -10 to -1 dB. The test set has 5×10^5 samples, with each SNR having 5×10^4 samples, that is to say, the RMSE corresponding to each SNR is averaged through 5×10^4 Monte Carlo experiments. The RMSE versus SNR curves for each method are shown in Fig. 5. From Fig. 5, it can be seen that the RMSE of ADMM-Net is basically consistent with ADMM and higher than IAA-APES. Although ADMM-Net has slightly lower RMSE than CVX and LMSL0 at low SNR, the computational complexity of the method in this letter is lower than that of the two aforementioned methods. A detailed analysis of the computational complexity can be found in Section IV-D. Considering the fewer iterations of ADMM-Net, the results indicate that ADMM-Net can achieve a better balance between estimation performance and efficiency.

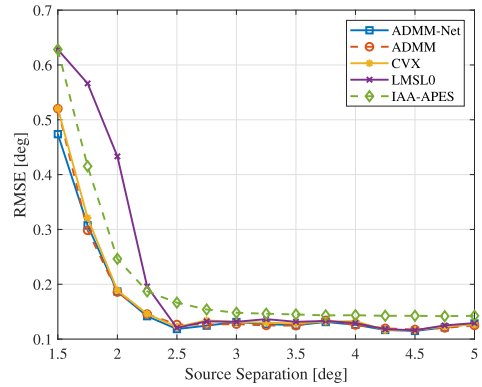

Fig. 7. RMSE performance with different source separation.

In addition, we compare the mean estimation error $\overline{\Delta\theta} = (\hat{\theta} - \theta)/L$, where L indicates the number of sample. The results for $\overline{\Delta\theta}$ are shown in Fig. 6. From the figure, we can see that the proposed method is an unbiased estimation method.

B. Performance Comparison of Source Separation

In this section, we compare the DOA estimation performance of different source separation. We set two sources with SNR $= -1$ dB. The source separation varies from 1.5° to 5°. It is worth noting that the bandwidth (BW) $= 1.8°$. We compare the ADMM-Net with IAA-APES, LMSL0, CVX, and ADMM in Fig. 7. From Fig. 7, we can see that the resolution capability of ADMM-Net is basically consistent with ADMM and CVX. LMSL0 exhibits larger RMSE at smaller source separations, indicating that the resolution capability of LMSL0 is weaker than that of ADMM-Net.

C. Three-Source DOA Estimation Performance Comparison

In this section, we compare the DOA estimation performance in the case of three sources with unequal powers. We set the DOAs of Sources 1 and 3 at 2.5° to the left and 3.5° to the right of Source 2. Also, the power of Sources 1 and 3 is 3 and 5 dB, respectively, below that of Source 2.

Fig. 8 shows the RMSEs of the three sources. The x-label of Fig. 8 demonstrates the SNR of Source 2. We see that for all Sources 1 and 2, the RMSE of ADMM-Net is basically consistent with ADMM and LMSL0, while for Source 3, the RMSE of LMSL0 is higher than that of ADMM-Net and ADMM. The results confirm that the ADMM-Net has great performance in DOA estimation with fewer iterations.

D. Run-Time Analysis

In this section, we analyze the run time of ADMM-Net and the methods used for performance comparison. The test set

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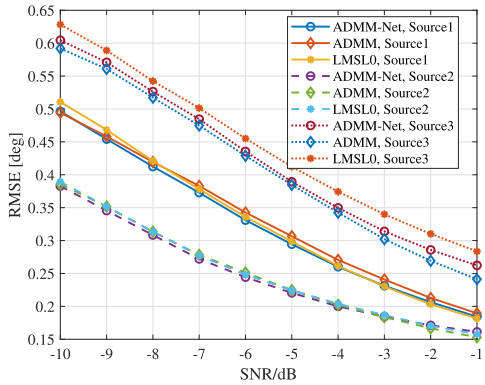

Fig. 8. RMSE performance for the case of three sources.

TABLE I
RUN TIME STATISTIC

Method	Run-time(s)
CVX	0.4567
IAA-APES	0.0910
LMSL0	0.0276
ADMM	0.0455
ADMM-Net	0.0117

in this section contains 5000 samples, with SNR uniformly distributed from -10 to -1 dB. We calculate the average run time per sample of each method. The result is shown in Table I. In CVX, semidefinite programming solver 3 (SDPT3) is used to solve the optimization problem. This solver has a high computational cost, leading to high run time per sample. IAA-APES requires the inverse of matrix, which results in a longer run time. LMSL0 is a two-level loop algorithm, so it has a longer run time. As for ADMM and ADMM-Net, the algorithms have the same computational cost in each iteration. However, according to our statistic, ADMM needs hundreds of iterations to converge, resulting in longer run time than ADMM-Net. The results in Table I indicate that the proposed ADMM-Net has computational efficiency. It can be seen that ADMM-Net achieves a better balance between DOA estimation performance and real-time constraints.

V. CONCLUSION

This letter proposes a novel DOA estimation method based on ADMM-Net. ADMM-Net is a deep architecture defined over a data flow graph determined by the ADMM algorithm, which is an improvement based on the idea of SR. Compared to the conventional ADMM algorithms, the ADMM-Net structure allows for independent parameter settings for each layer and fine-tunes the parameters through training, enabling ADMM-Net to achieve a better balance between DOA estimation performance and real-time constraints. However, the proposed method cannot adaptively adjust the number of iterations, which will be considered to improve in the future.

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